

MNC Overall Report

Technical Guide

Weather, logistics, livestock, wildlife, and patrol reporting for Mara North
Conservancy

Generated August 25, 2026

Workflow id: `mara_north_event_report`

1. Overview

The **mara_north_event_report** workflow (repository **mnc-overall-report**) is a comprehensive, multi-pipeline report for Mara North Conservancy. It pulls data from two independent sources — EarthRanger weather station observations and EarthRanger event records — and routes them through five reporting sections plus a cross-cutting events overview:

- **Section 1 — Weather:** Observations from the 'ER2ER - From GMMF' subject group; produces a daily summary CSV and 7 line charts (one per meteorological variable).
- **Section 2 — Logistics:** Balloon landings, airstrip operations, and airstrip maintenance events produce summary tables. A fourth event type, airline complaints, is fetched and normalised but produces no output (see Section 6.4).
- **Section 3 — Livestock:** Mobile boma movements, cattle counts, livestock predation, and illegal grazing events; produces summary CSVs/tables and maps that additionally show a grazing-zone layer not used elsewhere in the workflow.
- **Section 4 — Wildlife:** Five wildlife incident types and eight wildlife sighting species; produces CSVs, maps, bar charts, and summary tables, with elephant and buffalo receiving the richest treatment (herd composition + herd size).
- **Events overview:** A cross-section daily/typed event count and a total-events line chart, independent of the five sections above.
- **Section 5 — Patrol:** Foot, vehicle, and motorbike patrol trajectories derived from patrol_info events; produces patrol effort CSVs, a raw relocations file, three per-mode coverage maps, an overall coverage map, overall per-ranger efforts, and conservancy patrol occupancy.

Logistics, Livestock, Wildlife, and Patrol all share a single **events_temporal** DataFrame built from one **get_events** call (no filters applied at fetch time — each branch performs its own downstream filter_df on event_type). Weather uses a separate **get_subjectgroup_observations** call. The workflow concludes by generating a Word report from a Dropbox-hosted template and assembling a 44-widget results dashboard.

Output summary

Section	Output type	Key files
Weather	CSV + HTML/PNG charts	weather_summary_table.csv, 7 × *_readings_over_time.html/.png
Logistics	CSV (+ HTML)	balloon_landing_summary_table, airstrip_operations_summary_table, airstrip_maintenance_summary_table (no output for airline complaints)
Livestock	CSV/HTML + Map	mobile_boma_movement_summary_table.csv, total_cattle_count_summary_table, total_livestock_predation_summary_table.csv + livestock_predation_summary_table (csv/html pair with overlapping names, see 7.3), boma_movement_map, livestock_predation_events, illegal_grazing_map
Wildlife	CSV/HTML + Map + Chart	wildlife_events_recorded.csv, wildlife_incidents_* CSVs, wildlife_incidents_map, 8 species × (sightings map + overall__summary_table), elephant/buffalo additionally get a herd-size bar chart and a herd-types bubble map

Section	Output type	Key files
Events overview	CSV + Chart	total_events_recorded_by_date.csv, total_events_recorded_by_type.csv, total_events_recorded.html/.png
Patrol	CSV + GeoParquet + Map	patrol_events.csv, patrol_purpose_summary.csv, patrol_relocations.geoparquet, foot/vehicle/motorbike_patrol_efforts.csv, foot/vehicle/motor_patrol_map, overall_patrol_map, patrol_trajectories.geoparquet, overall_patrol_efforts.csv, patrol_coverage.csv
Report	Word document	overall_report.docx

2. Dependencies

2.1 Python packages

Package	Version	Channel
ecoscope-platform	>=2.15.0, <2.16.0	ecoscope-workflows
ecoscope-workflows-ext-custom	0.1.0rc14.*	ecoscope-workflows-custom
ecoscope-workflows-ext-ste	0.0.0rc1.*	ecoscope-workflows-custom
ecoscope-workflows-ext-wwf-virunga	0.0.0rc9.*	ecoscope-workflows-custom
ecoscope-workflows-ext-big-life	1.0.1.*	ecoscope-workflows-custom
ecoscope-workflows-ext-mnc	1.0.3.*	ecoscope-workflows-custom
pydeck	0.9.2	conda-forge
opentelemetry-sdk	>=1.20.0,<2.0.0	conda-forge

Several tasks in this spec call into the **ecoscope_workflows_ext_mnc.tasks.*** namespace: `fix_invalid_geometries`, `build_legend_values_from_column`, `remove_brackets_from_column`, `convert_columns_to_int`, and `apply_arithmetic_operation_over_rows`. The requirements list above pins **ecoscope-workflows-ext-mnc 1.0.3.***, which supplies these tasks — the package is present and correctly versioned in the current spec.

2.2 Connections and external assets

Asset	Task / Source	Purpose
EarthRanger	set_er_connection	Fetch all event records and subject group observations; also used by <code>process_events_details</code> calls to resolve display titles and by <code>get_patrol_values</code> / patrol observation fetches.
mnc_conservancy.gpkg	fetch_and_persist_file (Dropbox)	MNC community conservancy boundaries, one polygon per grazing_zone value. Fixed with <code>fix_invalid_geometries</code> , then split into the conservancy AOI, the Mara North extent, and the non-conservancy grazing zones (see Section 3).
mnc_across_the_river_parcel.gpkg	fetch_and_persist_file (Dropbox)	MNC across-the-river land parcels. Used as an additional polygon layer on every map in the workflow.
mara_north_event_template.docx	fetch_mnc_template / fetch_and_persist_file (Dropbox)	Word report template. Populated by <code>generate_mnc_report</code> at the end of the workflow to produce <code>overall_report.docx</code> .

Note: All Dropbox files are downloaded with `overwrite_existing: false` and `retries: 3`. If a file already exists in `ECOSCOPE_WORKFLOWS_RESULTS` from a prior run the download is skipped.

2.3 Grouper

The workflow uses an **empty grouper list** (`groupers: []`, set once by the `set_groupers` task and referenced via `workflow.groupers.return` `}}` everywhere a grouper is needed, including `add_temporal_index` calls and the final

dashboard). All records are processed as a single undivided dataset — no fan-out or per-group branching is applied anywhere in the spec.

3. Geospatial Asset Pipeline

Before the reporting sections run, the workflow downloads and prepares all shared geospatial base layers: two ArcGIS raster basemap tiles, the conservancy boundary / grazing-zone / parcels vector layers, and a single shared map zoom and view state. These are reused across every map drawn later in the workflow.

3.1 Conservancy boundaries

Step	Task	Detail
1	fetch_and_persist_file	Download mnc_conservancy.gpkg from Dropbox (overwrite_existing: false, retries: 3).
2	load_df	Load the gpkg into a GeoDataFrame.
3	fix_invalid_geometries	Repair invalid geometries (ecoscope_workflows_ext_mnc task). All three derived views below are built from this fixed layer.
4	filter_df — Conservancy AOI	grazing_zone == 'Conservancy' → the coverage/report area of interest, used as the base polygon for the patrol coverage grid and as one of the two/three layers combined on every map.
5	filter_df — Mara North extent	name == 'Mara North Conservancy' → used only to compute the shared map zoom/centre (Section 3.5), not drawn on any map.
6	filter_df — Grazing zones	grazing_zone != 'Conservancy' → the individual grazing zones, coloured with a GnBu colormap and a generated legend (build_legend_values_from_column, ecoscope_workflows_ext_mnc).

3.2 Styled zone layers

Three reusable static DeckGL layers are built, one per derived view above:

Layer	Task	Style	Legend
create_conservancy_layer	create_deckgl_layer_from_gdf (or equivalent)	Grey outline, unfilled	'Conservancy Boundary'
create_grazing_zones_layer	create_deckgl_layer_from_gdf	Filled per-zone colour from the GnBu legend	grazing_zone_legend_values
create_parcels_layer	create_deckgl_layer_from_gdf	Tan / khaki fill, stroked	'Parcels' (legend title: 'Map Layers')

Note: Layering rule: the three Livestock-branch maps (mobile boma movement, livestock predation, illegal grazing) combine **all three** static layers — parcels + grazing zones + conservancy. Every other map in the workflow (all wildlife sighting/incident maps and all patrol maps) combines only **two** layers — parcels + conservancy — the grazing-zone layer is intentionally omitted outside the Livestock section. This is a real, deliberate behavioural distinction and did not exist in the previous version of this spec.

3.3 Conservancy GDFs and text labels

Step	Task	Detail
1	create_gdf_from_dict / filter_df	Build conservancy_gdf (Conservancy AOI) and envelope_gdf (Mara North extent) for downstream use.
2	create_custom_text_layer	Render conservancy / zone name labels for placement on the maps.

3.4 Parcels layer

Step	Task	Detail
1	fetch_and_persist_file	Download mnc_across_the_river_parcelsgpkg from Dropbox.
2	load_df	Load the parcels gpkg into a GeoDataFrame.
3	create_parcel_layer	Render as a filled polygon layer (tan fill). Legend entry: 'Parcels', legend title 'Map Layers'.

3.5 Base tiles and shared view state

Item	Detail
Basemap tile layers	Two ArcGIS raster layers reused as tile_layers on every draw_map call: Elevation/World_Hillshade (opacity 0.80, max_zoom 20) and Reference/World_Boundaries_and_Places_Alternate (opacity 0.35, max_zoom 20).
Map Zoom & Extent task-group	compute_view_state_from_gdf on the Mara North extent (envelope_gdf) with pitch 0, bearing 0, max_zoom 15 → gdf_image_extent.

Note: gdf_image_extent is the single view_state reused on every draw_map call in the entire workflow — Livestock, Wildlife, and Patrol maps alike share one zoom/centre, computed once.

4. Shared Event Ingestion Pipeline

Logistics, Livestock, Wildlife, and Patrol all share a single event retrieval call. Weather data uses a separate subject-group observation fetch (Section 5.1).

4.1 Event retrieval

Parameter	Value
Task	get_events
event_types	[] (fetch all event types — no filter at fetch time)
Downstream filtering	Each of Logistics, Livestock, Wildlife Incidents, Wildlife Sightings, Events Overview, and Patrol performs its own filter_df / exclude_row_values on event_type against the shared events_temporal DataFrame.
include_details	true
raise_on_empty	true
include_null_geometry	false

4.2 Date extraction and temporal indexing

Step	Task	Detail
1	extract_column_as_type	Extract the time column as output_type: date into a new column named date .
2	add_temporal_index	Add temporal index using the shared groupers (\${{ workflow.groupers.return }}, i.e. []). Produces the shared events_temporal DataFrame consumed by every downstream section.

4.3 Common event detail normalisation pattern

Every event branch applies the same three-step normalisation after filtering events_temporal by event_type:

Step	Task	Detail
1	process_events_details	Resolve event detail field IDs to display titles (map_to_titles: true). Requires the ER connection.
2	normalize_json_column	Flatten the event_details JSON column.
3	drop_column_prefix	Remove the event_details__ prefix from all flattened columns.

Note: Because map_to_titles is true, flattened column names are human-readable display titles from EarthRanger (e.g. 'Balloon Company', 'Livestock Species'). All downstream map_columns steps reference these titles directly.

5. Section 1 — Weather Pipeline

The weather pipeline runs independently from the shared event pipeline. It fetches sensor observations from the **ER2ER - From GMMF** subject group and extracts seven meteorological fields sequentially from the **extra__observation_details** JSON column.

5.1 Observation retrieval

Parameter	Value
Task	get_subjectgroup_observations
subject_group_name	ER2ER - From GMMF

5.2 Field extraction chain

Seven sequential **extract_value_from_json_column** tasks are chained, each reading from the previous task's output, all extracting from **extra__observation_details**:

Output column	Field
precipitation	precipitation
temperature	temperature
wind_speed	wind_speed
wind_gusts	wind_gusts
soil_temperature	soil_temperature
relative_humidity	relative_humidity
atmospheric_pressure	atmospheric_pressure

5.3 Date extraction, renaming, and temporal index

Step	Task	Detail
1	extract_column_as_type	Extract the observation time column as output_type: date → new column date .
2	map_columns	Rename extra__subject__name → weather_station .
3	add_temporal_index	Add temporal index (groupers: []).

5.4 Daily weather summary

Task: **summarize_df**. Groups by **weather_station** and **date**. Aggregations:

Column	Aggregator	Description
precipitation	sum	Total daily rainfall

Column	Aggregator	Description
temperature	mean	Daily average temperature
wind_speed	mean	Daily average wind speed
wind_gusts	max	Daily maximum wind gust
soil_temperature	mean	Daily average soil temperature
relative_humidity	mean	Daily average relative humidity
atmospheric_pressure	mean	Daily average atmospheric pressure

Output saved as **weather_summary_table.csv**. No dashboard widget is created for this table — it persists as a downloadable file only.

5.5 Line charts

One **draw_line_chart** task per metric reads from the daily weather summary (x_column: date, category_column: weather_station), each persisted as HTML then converted to PNG:

HTML filename	Widget
precipitation_readings_over_time.html	Precipitation
temperature_readings_over_time.html	Temperature
wind_speed_readings_over_time.html	Wind Speed
wind_gusts_readings_over_time.html	Wind Gusts
soil_temperature_readings_over_time.html	Soil Temperature
relative_humidity_readings_over_time.html	Relative Humidity
atmospheric_pressure_readings_over_time.html	Atmospheric Pressure

Note: All 7 weather widgets are plot widgets on the dashboard (see Section 12), and all are widget-creation steps — they use skipif.conditions: [never], so a placeholder still renders even if the weather observation fetch returned no data.

6. Section 2 — Logistics Report

Four event types are filtered from **events_temporal** and normalised using the common three-step pattern (Section 4.3): **balloon_landing**, **airstrip_operations**, **airstrip_maintenance**, and **airline_complaint**.

6.1 Branch 1 — Balloon Landings

Filter: `event_type == balloon_landing`

Step	Task	Detail
1–3	normalise	<code>process_events_details</code> → <code>normalize_json_column</code> → <code>drop_column_prefix</code> .
4	<code>map_columns</code>	Retain <code>date</code> , <code>Balloon Company</code> , <code>Where are clients staying?</code> , <code># of passengers</code> ; rename for display.
5	<code>remove_brackets_from_column</code>	Strip list-bracket characters from the retained text columns (<code>ecoscope_workflows_ext_mnc</code> task).
6	<code>persist_df</code>	Save as balloon_landing_summary_table.csv .
7	<code>draw_table</code> + <code>persist_text</code>	Render an HTML table; persisted with a runtime-computed <code>filename_suffix</code> of <code>'balloon_landing_summary_table.html'</code> (there is no explicit <code>`filename:`</code> for the HTML output — the name is derived at run time from the suffix).

Widget: **'Balloon Landing Summary'** (table).

6.2 Branch 2 — Airstrip Operations

Filter: `event_type == airstrip_operations`

Step	Task	Detail
1–3	normalise	<code>process_events_details</code> → <code>normalize_json_column</code> → <code>drop_column_prefix</code> .
4	<code>summarize_df</code>	Group by <code>camp/lodge</code> × <code>arrival/departure</code> ; count clients.
5	<code>pivot_df</code>	Pivot the grouped counts into <code>Arrival / Departure</code> columns, one row per <code>Camp/Lodge</code> .
6	<code>persist_df</code>	Save as airstrip_operations_summary_table.csv .
7	<code>draw_table</code> + <code>persist_text</code>	Render an HTML table; persisted with a runtime-computed <code>filename_suffix</code> of <code>'airstrip_operations_summary_table.html'</code> .

Widget: **'Airstrip Operations Summary'** (table).

6.3 Branch 3 — Airstrip Maintenance

Filter: `event_type == airstrip_maintenance`

Step	Task	Detail
1–3	normalise	<code>process_events_details</code> → <code>normalize_json_column</code> → <code>drop_column_prefix</code> .

Step	Task	Detail
4	map_columns	Retain date and Maintenance type.
5	persist_df	Save as airstrip_maintenance_summary_table.csv .
6	draw_table + persist_text	Render an HTML table; persisted with a runtime-computed filename_suffix of 'airstrip_maintenance_summary_table.html'.

Widget: **'Airstrip Maintenance Summary'** (table).

6.4 Branch 4 — Airline Complaints

Filter: event_type == **airline_complaint**.

Note: This branch is fetched, filtered, and normalised (process_events_details → normalize_json_column → drop_column_prefix) exactly like the other three, but the pipeline **dead-ends there**: there is no map_columns, persist_df, draw_table, or widget-creation step downstream of the normalisation for airline complaints anywhere in the spec. **No file and no dashboard widget are produced for airline complaints** in the current version of this workflow. This is a real behavioural difference from earlier versions and is worth flagging explicitly to anyone expecting an airline-complaints output.

7. Section 3 — Livestock Report

Four livestock event types are filtered from **events_temporal** and each normalised using the common pattern (Section 4.3). All Livestock maps use the 3-layer map style (parcels + grazing zones + conservancy) described in Section 3.2 — the only maps in the workflow that include the grazing-zone layer.

7.1 Branch 1 — Mobile Boma Movements

Filter: event_type == **mobile_boma_rep**

Step	Task	Detail
1–3	normalise	process_events_details → normalize_json_column → drop_column_prefix.
4	map_columns	Retain id, date, event_type, geometry, Date of Relocation, Electric Boma Status, Mobile Boma Zone, Nature of the Site, Reason for relocation.
5	summarize_df	Group by date; count nunique(id) boma events.
6	persist_df	Save as mobile_boma_movement_summary_table.csv .
7	draw_map	Point map, coloured layer combined with the 3-layer livestock map style; persisted as boma_movement_map.html/png .

Widget: '**Mobile Boma Movement Map**' (map).

7.2 Branch 2 — Total Cattle Count

Filter: event_type == **cattle_count**. No map produced.

Step	Task	Detail
1–3	normalise	process_events_details → normalize_json_column → drop_column_prefix.
4	map_columns	Rename '# cattle in Zone 1 mobile boma' → zone_1, '# cattle in Zone 2/3 mobile boma' → zone_2_3, '# cattle in Zone 4' → zone_4.
5	apply_arithmetic_operation_over_rows	Sum zone_1 + zone_2_3 + zone_4 into a new total column (ecoscope_workflows_ext_mnc task).
6	persist_df	Save as total_cattle_count_summary_table.csv .
7	draw_table + persist_text	Save the rendered HTML with an explicit filename total_cattle_count_summary_table.html (filename_suffix: null — unlike the Logistics branches, this one names its HTML output directly).

Widget: '**Total Cattle Count Summary**' (table).

7.3 Branch 3 — Livestock Predation

Filter: event_type == **livestock_predation_rep**. This branch builds two genuinely different output tables that share an overlapping naming scheme — read carefully:

Output file	Content	Backs a widget?
total_livestock_predation_summary_table.csv	Daily nunique-id predation event counts. Columns: 'Date', 'Livestock Predation Events'.	No (CSV only)
livestock_predation_summary_table.html	The rendered HTML of that same daily-count table above (not a different dataset).	Yes — 'Livestock Predation Summary'
livestock_predation_summary_table.csv	A second, different table: the detailed per-event records (Livestock Species, Suspected Predator, Total livestock affected), with nulls filled 'Unknown'.	No (CSV only, no HTML/widget)
livestock_predation_events.html/.png	Point map of predation incidents, coloured by species (tab10 colormap).	Yes — 'Livestock Predation Events Map'

Note: The daily-count CSV (total_livestock_predation_summary_table.csv) and the detailed per-event CSV (livestock_predation_summary_table.csv) have similar but distinct names and different columns entirely. The 'Livestock Predation Summary' dashboard widget is backed by the HTML of the **daily-count** table, not the detailed per-event table — the detailed CSV has no widget at all.

7.4 Branch 4 — Illegal Grazing

Filter: event_type == **illegal_grazing_rep**. Only a map is produced — no summary CSV or table exists for this branch.

Step	Task	Detail
1–3	normalise	process_events_details → normalize_json_column → drop_column_prefix.
4	map_columns	Retain date, event_type, geometry, Herd Zone, Landowner name, action taken.
5	draw_map	Point map on the 3-layer livestock map style; persisted as illegal_grazing_map.html/.png .

Widget: '**Illegal Grazing Events Map**' (map).

8. Section 4 — Wildlife Report

This section covers five wildlife incident types (fetched together as one branch) and eight wildlife sighting species (each its own branch). All Wildlife maps use the 2-layer map style (parcels + conservancy only) described in Section 3.2 — the grazing-zone layer used in Livestock maps is intentionally absent here.

8.1 Common wildlife sighting pattern

Every one of the 8 sighting branches follows the same skeleton:

Step	Detail
1	filter_df events_temporal by event_type
2–4	process_events_details → normalize_json_column → drop_column_prefix
5	map_columns to rename species-specific fields to snake_case
6	exclude_geom_outliers → drop_null_geometry
7	draw_map on the shared parcels + conservancy (2-layer) map style

Branches differ in sophistication across three tiers, described in the branch list below (Section 8.2) and detailed further in Sections 8.3–8.4.

8.2 Sighting branches and daily count / summary CSV outputs

Species	event_type value	Tier
Elephant	elephant_sighting_rep	Full (herd composition + herd size)
Buffalo	buffalo_sighting_rep	Full (herd composition + herd size)
Lion	lion_sighting_rep	Mid (pride, map + summary)
Leopard	leopardsightingrep	Mid — NOTE: event_type has no underscores
Cheetah	cheetah_sighting_rep	Mid (map + summary)
Giraffe	giraffe_sighting	Simplest — NOTE: no '_rep' suffix
Rhino	rhino_sighting_rep	Simplest (map + summary)
Hartebeest	hartebeest_sighting	Simplest — NOTE: no '_rep' suffix

Note: Leopard's event_type is written verbatim as 'leopardsightingrep' (no underscores at all), while Giraffe and Hartebeest use 'giraffe_sighting' / 'hartebeest_sighting' (underscored, but missing the '_rep' suffix that every other species carries). These are exact values from the spec, not typos to correct.

8.3 Individual / pride summary CSVs

Species	Outputs (CSV + HTML)	Widgets
Elephant	elephant_herd_size_bar_chart.html/.png, elephant_herd_types_map.html/.png, elephant_sightings_events.html/.png, overall_elephant_summary_table.csv + .html	Elephant Herd Size Map, Elephant Herd Composition Map, Elephant Herd Size Distribution, Elephant Herd Composition Summary
Buffalo	buffalo_herd_size_bar_chart.html/.png, buffalo_herd_types_map.html/.png, buffalo_sightings_events.html/.png, overall_buffalo_summary_table.csv + .html	Buffalo Herd Size Map, Buffalo Herd Composition Map, Buffalo Herd Size Distribution, Buffalo Herd Composition Summary
Lion	lion_pride_sightings_map.html/.png, overall_lion_summary_table.csv + .html	Lion Sightings Map, Lion Sightings Summary
Leopard	leopard_sightings_map.html/.png, overall_leopard_summary_table.csv + .html	Leopard Sightings Map, Leopard Sightings Summary
Cheetah	cheetah_sightings_map.html/.png, overall_cheetah_summary_table.csv + .html	Cheetah Sightings Map, Cheetah Sightings Summary
Giraffe	giraffe_sightings_map.html/.png, overall_giraffe_summary_table.csv + .html	Giraffe Sightings Map, Giraffe Sightings Summary
Rhino	rhino_sightings_map.html/.png, overall_rhino_summary_table.csv + .html	Rhino Sightings Map, Rhino Sightings Summary
Hartebeest	hartebeest_sightings_map.html/.png, overall_hart_summary_table.csv + .html	Hartebeest Sightings Map, Hartebeest Sightings Summary

Note: Hartebeest's summary CSV/HTML filenames use the abbreviation 'hart' — **overall_hart_summary_table.csv/.html** — not 'hartebeest'. This is inconsistent with the map filename (hartebeest_sightings_map.html) and with the widget titles ('Hartebeest Sightings Map/Summary'), but is the exact filename produced by the spec.

8.4 Sighting maps

Elephant and Buffalo (full tier): herd_composition is mapped to Bachelor / Female+Calf / Mixed / Unspecified and drives the colour of the herd-composition map (elephant_sightings_events / buffalo_sightings_events); herd_size is binned (equal-interval, $k = 5$) and drives a bar chart (*_herd_size_bar_chart) plus a separate clustered 'herd size' bubble map (*_herd_types_map).

Lion, Leopard, Cheetah (mid tier): a single point map coloured by pride (Lion) or individuals present (Leopard, Cheetah), plus a single aggregate summary table — no bar chart or bubble map.

Giraffe, Rhino, Hartebeest (simplest tier): a fixed-colour point map with no attribute breakdown; the summary table is a plain daily observation count.

8.5 Elephant/buffalo herd-size map and bar chart

Output	Detail
Herd-size bar chart	herd_size binned (equal-interval, $k = 5$) → draw_bar_chart → persist HTML → html_to_png. Files: elephant_herd_size_bar_chart / buffalo_herd_size_bar_chart (.html/.png).

Output	Detail
Herd-size bubble map	A clustered scatterplot layer sized by herd_size, combined with the 2-layer conservancy+parcels map style → draw_map → persist HTML → html_to_png. Files: elephant_herd_types_map / buffalo_herd_types_map (.html/.png).
Herd-composition map	A separate point map coloured by herd_composition. Files: elephant_sightings_events / buffalo_sightings_events (.html/.png).

8.6 Wildlife incidents

Filter: event_type in {snare_rep, fire_rep, wildlife_injury_rep, wildlife_treatment_rep, wildlife_carcass_rep}. Raw records are persisted, a categorical summary and a daily-count table are built, and a point map coloured by event type is drawn using the **ecoscope_workflows_ext_big_life** draw_map task — functionally equivalent to the custom draw_map task used by Livestock, but a different implementation module.

Output file	Description	Widget
wildlife_events_recorded.csv	Raw normalised wildlife incident records.	None (CSV only)
wildlife_incidents_summary_table.csv	Categorical incident summary by event type.	None (CSV only)
wildlife_incidents_recorded_by_date.csv	Daily incident counts.	None (CSV only)
wildlife_incidents_map.html/.png	Point map of incident locations, coloured by event type.	'Wildlife Incident Map'

9. Section 5 — Patrol Report

The patrol section derives trajectory data from **patrol_info** events in `events_temporal`, then fetches full patrol records and GPS observations from EarthRanger. It produces event counts, a patrol purpose summary, three mode-specific coverage maps, an overall coverage map, per-mode and per-ranger effort summaries, and a conservancy patrol occupancy table. Section 9.1 below also covers the cross-section 'Events Overview' outputs, which sit alongside the patrol-specific pipeline but are not themselves patrol data.

9.1 Event counts (Events Overview)

Before the patrol-specific pipeline, a cross-section events overview is built from `events_temporal` with **distancecountwildlife_rep**, **distancecountpatrol_rep**, and **airstrip_operations** excluded (`exclude_row_values`):

CSV filename	Group by	Description
<code>total_events_recorded_by_date.csv</code>	<code>date</code>	Unique event count per date
<code>total_events_recorded_by_type.csv</code>	<code>date + event_type</code>	Unique event count per date per event type

A line chart of total events per day is also drawn and saved as **total_events_recorded.html/png**. Widget: 'Total Events Recorded' (plot).

9.2 Patrol purpose summary

Filter: `events_temporal` for `event_type == patrol_info`. Records are processed/normalised and the raw filtered records are persisted as **patrol_events.csv** (no widget). The events are then summarised by **patrol_purpose**:

Output file	Description	Widget
<code>patrol_events.csv</code>	Raw normalised <code>patrol_info</code> event records.	None (CSV only)
<code>patrol_purpose_summary.csv</code>	Patrol counts summarised by <code>patrol_purpose</code> .	None (CSV only)
<code>patrol_purpose_summary_table.html</code>	Rendered HTML of the purpose summary.	'Patrol Purpose Summary'

9.3 Patrol observations and relocations

Step	Task	Detail
1	<code>explode_multiple_columns</code>	Explode <code>patrol_id</code> (and participant) lists into rows.
2	<code>get_patrol_values</code>	Fetch full patrol records from EarthRanger by <code>patrol_id</code> .
3	<code>get patrol observations</code>	Fetch GPS observations for each patrol.
4	<code>merge / explode participants</code>	Merge observations with patrol info; explode participants.
5	<code>process_relocations</code>	Convert observations to relocations.
6	<code>persist_df</code>	Save the raw, pre-mode-split relocation points as patrol_relocations.geoparquet .

9.4 Trajectory building

Relocations are split by **transport_type** into Foot, Vehicle, and Motorbike sub-groups, each independently converted to trajectories with mode-specific segment filters. Foot patrols use much tighter speed/time thresholds than vehicle or motorbike patrols, reflecting the different realistic travel speeds:

Sub-group	transport_type filter	Segment filter characteristics
Foot patrols	Foot	Tight speed/time thresholds appropriate to walking pace
Vehicle patrols	Vehicle	Wider speed/time thresholds appropriate to vehicle travel
Motorbike patrols	Motorbike	Wider speed/time thresholds appropriate to motorbike travel

9.5 Patrol effort CSVs

CSV filename	Group-by	Columns
foot_patrol_efforts.csv	patrol_type_value	no_of_patrols, distance_km, duration_hrs, average_speed
vehicle_patrol_efforts.csv	patrol_type_value	no_of_patrols, distance_km, duration_hrs, average_speed
motorbike_patrol_efforts.csv	patrol_type_value	no_of_patrols, distance_km, duration_hrs, average_speed
overall_patrol_efforts.csv	participants (i.e. per-ranger)	number_of_patrols, distance_km, duration_hours (no average_speed column — a different shape from the three per-mode tables above)

Note: overall_patrol_efforts.csv is **not** simply the three per-mode tables concatenated — it re-summarises the combined trajectories grouped by **participants** (ranger), producing a per-ranger effort breakdown rather than a per-patrol-type one.

9.6 Patrol trajectory maps

Each mode produces its own coverage grid and map: build a 1 km grid over the conservancy boundary, classify unique patrol visits per cell (equal-interval, $k = 5$), colour with RdYlGn, and render via draw_map:

Mode	Effort CSV	Map filename	Widget
Foot	foot_patrol_efforts.csv	foot_patrol_map.html/.png	Foot Patrol Coverage Map
Vehicle	vehicle_patrol_efforts.csv	vehicle_patrol_map.html/.png	Vehicle Patrol Coverage Map
Motorbike	motorbike_patrol_efforts.csv	motor_patrol_map.html/.png	Motorbike Patrol Coverage Map

Note: The motorbike map's **filename** uses the abbreviation 'motor' — **motor_patrol_map.html/.png** — while its **widget title** reads 'Motorbike Patrol Coverage Map' in full. This filename/title mismatch is exact-as-specified, not an error to fix in this guide.

9.7 Overall patrol coverage map and occupancy

Foot, vehicle, and motorbike trajectories are concatenated and a combined coverage grid/map is built the same way as the per-mode maps. This section produces two **distinct** kinds of output — a visual coverage map and a numeric occupancy table — which should not be conflated:

Output file	Nature	Widget
overall_patrol_map.html/.png	The overall coverage MAP (visual) — combined foot+vehicle+motor coverage grid rendered spatially. This is the only 'overall coverage map' output in the workflow.	'Overall Patrol Coverage Map'
patrol_trajectories.geoparquet	Despite the task name ('Persist combined trajectories data') and the filename, this file actually persists the reprojected overall coverage grid (density cells), not raw trajectory line geometries. Confirmed against the spec: it is written from the same reprojected grid DataFrame used to build overall_patrol_map.	None (file only)
overall_patrol_efforts.csv + overall_patrol_efforts_table.html	Per-ranger effort summary (see Section 9.5).	'Overall Patrol Efforts'
patrol_coverage.csv + patrol_coverage_table.html	Conservancy occupancy percentage from compute_patrol_occupancy — a plain numeric TABLE, not a map.	'Conservancy Patrol Occupancy'

Note: There is no separate 'patrol coverage map' distinct from a 'patrol coverage table' the way earlier documentation implied. The visual coverage-map role is filled entirely by **overall_patrol_map.html**; patrol_coverage_table.html is a plain occupancy-percentage table widget with no spatial rendering at all.

10. Report Generation

The workflow concludes with two final-assembly tasks: a Word report and a results dashboard.

10.1 Word report

Parameter	Value
Task id	generate_overall_report
Task ref	generate_mnc_report
template_path	mara_north_event_template.docx, fetched by fetch_mnc_template (Dropbox)
output_dir	ECOSCOPE_WORKFLOWS_RESULTS
generated_by	Ecoscope
validate_images	true
time_period	from time_range
filename	overall_report.docx

Note: Unlike the dashboard's `gather_dashboard` task, `generate_overall_report`'s partial does **not** enumerate any explicit list of images, tables, or sections in `spec.yaml`. The order and content of the assembled document is driven entirely by the structure of the Word template itself (`mara_north_event_template.docx`) and whatever `generate_mnc_report`'s implementation scans from `output_dir` at run time — it is **not** something declared or orderable in `spec.yaml`. This guide does not assert a section order for the docx because none is specified.

10.2 Dashboard

Task id **mnc_events_dashboard**, task ref **gather_dashboard**, built from `workflow_details`, `time_range`, and groupers plus an explicit ordered **widgets**: list of 44 widgets, grouped (by spec comments) in this exact order:

Group	Count	Widgets
Weather	7	Precipitation, Temperature, Wind Speed, Wind Gusts, Soil Temperature, Relative Humidity, Atmospheric Pressure
Logistics	3	Balloon Landing Summary, Airstrip Operations Summary, Airstrip Maintenance Summary
Livestock	5	Mobile Boma Movement Map, Livestock Predation Events Map, Illegal Grazing Events Map, Total Cattle Count Summary, Livestock Predation Summary
Wildlife	21	Wildlife Incident Map; Elephant Herd Size Map / Herd Composition Map / Herd Size Distribution / Herd Composition Summary; Buffalo (same 4); Lion Sightings Map / Summary; Leopard Sightings Map / Summary; Cheetah Sightings Map / Summary; Giraffe Sightings Map / Summary; Hartebeest Sightings Map / Summary; Rhino Sightings Map / Summary
Patrol	8	Foot / Vehicle / Motorbike / Overall Patrol Coverage Map, Total Events Recorded, Patrol Purpose Summary, Overall Patrol Efforts, Conservancy Patrol Occupancy

Note: No dashboard widget exists for: weather_summary_table, wildlife_events_recorded.csv / wildlife_incidents_summary_table.csv / wildlife_incidents_recorded_by_date.csv, total_events_recorded_by_date.csv / total_events_recorded_by_type.csv, patrol_events.csv, patrol_relocations.geoparquet, and patrol_trajectories.geoparquet — these persist as downloadable files only and never surface on the dashboard.

11. Output Files

All outputs are written to **ECOSCOPE_WORKFLOWS_RESULTS**.

11.1 Weather

File	Description
weather_summary_table.csv	Daily per-station summary: precipitation (sum), temperature (mean), wind speed (mean), wind gusts (max), soil temperature (mean), relative humidity (mean), atmospheric pressure (mean)
precipitation_readings_over_time.html/.png	Precipitation line chart per weather station
temperature_readings_over_time.html/.png	Temperature line chart per weather station
wind_speed_readings_over_time.html/.png	Wind speed line chart
wind_gusts_readings_over_time.html/.png	Wind gusts line chart
soil_temperature_readings_over_time.html/.png	Soil temperature line chart
relative_humidity_readings_over_time.html/.png	Relative humidity line chart
atmospheric_pressure_readings_over_time.html/.png	Atmospheric pressure line chart

11.2 Logistics

File	Description
balloon_landing_summary_table.csv (+ HTML)	Balloon company, lodging, and passenger records
airstrip_operations_summary_table.csv (+ HTML)	Client counts per camp/lodge, pivoted by arrival/departure
airstrip_maintenance_summary_table.csv (+ HTML)	Dated log of airstrip maintenance activity types
— (none)	Airline complaints are fetched and normalised but produce no output file or widget (see Section 6.4)

11.3 Livestock

File	Description
mobile_boma_movement_summary_table.csv	Daily unique boma event count
boma_movement_map.html/.png	Mobile boma locations, 3-layer map (parcels + grazing zones + conservancy)

File	Description
total_cattle_count_summary_table.csv (+ HTML)	Cattle counts per zone (zone_1, zone_2_3, zone_4) plus a computed total column
total_livestock_predation_summary_table.csv	Daily unique predation event count
livestock_predation_summary_table.html	Rendered HTML of the daily predation count table (see 7.3)
livestock_predation_summary_table.csv	Separate detailed per-event predation table (species, predator, total affected) — same filename stem as the HTML above but a different dataset (see 7.3)
livestock_predation_events.html/.png	Predation incident locations, coloured by species (tab10)
illegal_grazing_map.html/.png	Illegal grazing incidents, 3-layer map (no summary CSV/table exists)

11.4 Wildlife

File	Description
wildlife_events_recorded.csv	Raw wildlife incident records (snares, fires, carcasses, injuries, veterinary treatments)
wildlife_incidents_summary_table.csv	Incident counts by event type
wildlife_incidents_recorded_by_date.csv	Incident counts by date
wildlife_incidents_map.html/.png	Wildlife incident locations map, coloured by event type
elephant_sightings_events.html/.png	Elephant herd-composition sightings map
elephant_herd_types_map.html/.png	Elephant herd-size bubble map
elephant_herd_size_bar_chart.html/.png	Elephant herd-size distribution bar chart (5 bins)
overall_elephant_summary_table.csv (+ HTML)	Elephant sightings summary by herd type
buffalo_sightings_events.html/.png	Buffalo herd-composition sightings map
buffalo_herd_types_map.html/.png	Buffalo herd-size bubble map
buffalo_herd_size_bar_chart.html/.png	Buffalo herd-size distribution bar chart (5 bins)
overall_buffalo_summary_table.csv (+ HTML)	Buffalo sightings summary by herd type
lion_pride_sightings_map.html/.png	Lion sightings map, coloured by pride
overall_lion_summary_table.csv (+ HTML)	Lion sightings summary by pride
leopard_sightings_map.html/.png	Leopard sightings map, coloured by individuals present
overall_leopard_summary_table.csv (+ HTML)	Leopard sightings summary

File	Description
cheetah_sightings_map.html/.png	Cheetah sightings map, coloured by individuals present
overall_cheetah_summary_table.csv (+ HTML)	Cheetah sightings summary
giraffe_sightings_map.html/.png	Giraffe sightings map (fixed colour)
overall_giraffe_summary_table.csv (+ HTML)	Giraffe daily sightings summary
rhino_sightings_map.html/.png	Rhino sightings map (fixed colour)
overall_rhino_summary_table.csv (+ HTML)	Rhino daily sightings summary
hartebeest_sightings_map.html/.png	Hartebeest sightings map (fixed colour)
overall_hart_summary_table.csv (+ HTML)	Hartebeest daily sightings summary — NOTE filename uses abbreviation 'hart', not 'hartebeest'

11.5 Patrol

File	Description
patrol_events.csv	Raw normalised patrol_info event records
patrol_purpose_summary.csv (+ HTML)	Patrol count and distance by purpose
patrol_relocations.geoparquet	Raw, pre-mode-split GPS relocations for all patrol types
foot_patrol_efforts.csv	Foot patrol metrics (patrols, distance, duration, avg speed) by patrol_type_value
foot_patrol_map.html/.png	Foot patrol coverage grid map (1 km cells, RdYIGn)
vehicle_patrol_efforts.csv	Vehicle patrol metrics by patrol_type_value
vehicle_patrol_map.html/.png	Vehicle patrol coverage grid map
motorbike_patrol_efforts.csv	Motorbike patrol metrics by patrol_type_value
motor_patrol_map.html/.png	Motorbike patrol coverage grid map — NOTE filename uses 'motor', widget title says 'Motorbike'
overall_patrol_map.html/.png	Combined (foot + vehicle + motorbike) patrol coverage grid map
patrol_trajectories.geoparquet	NOTE: despite the name, this is the reprojected overall coverage GRID (density cells), not raw trajectory line geometries — see Section 9.7
overall_patrol_efforts.csv (+ HTML)	Per-ranger (participants) summary of patrols, distance, and duration
patrol_coverage.csv (+ HTML)	Conservancy patrol occupancy percentage — a numeric table, not a map

11.6 Report

File	Description
mara_north_event_template.docx	Word template downloaded from Dropbox (input, not an output)
overall_report.docx	Populated Word report; section order is template-driven, not declared in spec.yaml (see Section 10.1)

12. Workflow Execution Logic

12.1 Per-task skip conditions

This workflow declares a global **task-instance-defaults** block applying the same skipif conditions to every task:

Condition	Behaviour
any_is_empty_df	Skip this task if any input DataFrame is empty
any_dependency_skipped	Skip this task if any upstream dependency was skipped

Most individual tasks additionally redeclare this same skipif block explicitly. **Widget-creation tasks** (e.g. `create_table_widget_single_view`, the map/plot widget builders) override the default with **skipif.conditions: [never]**, so a placeholder widget always renders on the dashboard even if the upstream branch that would feed it was skipped for lack of data.

Note: Because skips propagate per-task rather than being suppressed globally, each section's branches skip independently. For example, if no balloon_landing events are returned, only the balloon branch skips; all other Logistics/Livestock/Wildlife/Patrol branches continue normally, and the corresponding widgets still render (empty) due to the [never] override.

12.2 Two independent data sources

Entry point	Task	Feeds
Weather observations	<code>get_subjectgroup_observations</code> (ER2ER - From GMMF)	Section 1 — Weather only
All event records	<code>get_events</code> (event_types: [], fetch all)	Sections 2–5 — Logistics, Livestock, Wildlife, Events Overview, Patrol (each applies its own downstream filter)

12.3 No mapvalues or fan-out

This workflow processes all records as a single batch. There is no **mapvalues**, **split_groups**, or **zip_groupbykey** directive, and groupers is []; every task runs exactly once per workflow execution.

12.4 GeoJSON trajectories and URL rewriting

Patrol trajectory / coverage maps that render vector overlays use a geospatial layer referencing an on-disk file. Before HTML-to-PNG conversion, a URL-rewriting step replaces the `file://` reference in the rendered HTML with a locally-served path so the headless browser used for screenshotting can load the geometry.

12.5 HTML-to-PNG conversion settings

Every HTML chart/map output in the workflow is converted to PNG via `html_to_png`, with settings tuned per output type — fast, low-wait conversion for simple charts, longer wait times for map renders that need basemap tiles and vector layers to finish loading before the screenshot is taken.

12.6 Known quirks (summary)

The following naming/behaviour discrepancies were identified while auditing this spec and are called out individually in their respective sections; they are collected here for quick reference:

- **Airline complaints dead-end** (6.4): fetched and normalised but no file or widget is ever produced.
- **Livestock predation dual naming** (7.3): `livestock_predation_summary_table.html` backs the daily-count widget, while the identically-stemmed `livestock_predation_summary_table.csv` is a completely different, detailed per-event dataset.
- **Hartebeest 'hart' abbreviation** (8.3): `overall_hart_summary_table.csv/.html`, inconsistent with the map filename and widget titles which say 'Hartebeest' in full.
- **Motorbike filename vs. widget title** (9.6): `motor_patrol_map.html/.png` backs the 'Motorbike Patrol Coverage Map' widget.
- **patrol_trajectories.geoparquet content mismatch** (9.7): named and task-labelled as combined trajectories, but actually persists the reprojected overall coverage grid (density cells).
- **Non-underscored / short event_type values** (8.2): `leopardsightingrep` (no underscores), `giraffe_sighting` and `hartebeest_sighting` (no `'_rep'` suffix), unlike every other sighting/incident `event_type` in the spec.
- **Report assembly order is template-driven** (10.1): unlike the dashboard, `spec.yaml` declares no explicit section/image list for the Word report — order comes entirely from `mara_north_event_template.docx`.

13. Software Versions

Package	Version pinned in spec.yaml
ecoscope-platform	>=2.15.0, <2.16.0
ecoscope-workflows-ext-custom	0.1.0rc14.*
ecoscope-workflows-ext-ste	0.0.0rc1.*
ecoscope-workflows-ext-wwf-virunga	0.0.0rc9.*
ecoscope-workflows-ext-big-life	1.0.1.*
ecoscope-workflows-ext-mnc	1.0.3.*
pydeck	0.9.2
opentelemetry-sdk	>=1.20.0,<2.0.0